

# POLICEPROOF

From Complaint to Resolution

Transparent • Smart • Accountable

Innovation & Product Idea Assignment

A practical technology-based solution for organized police case tracking  
and evidence management

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## Project Type: Option B — New Product

PoliceProof is a completely new software product that solves a real-world public-service problem. It can be developed as an MVP by focusing on the minimum features needed for complaint and case tracking.

# 1. Problem Statement

## What problem are you trying to solve?

After a citizen reports a crime, the journey from complaint to investigation can be difficult to understand and organize. Police personnel may also need a structured way to manage case information, assignments, investigation updates, evidence activity, and supervisory review in one place.

## Who experiences this problem?

- Citizens who submit complaints and want to know the permitted status of their case.
- Police officers who receive and manage complaints and need an organized workflow.
- Investigators who handle evidence and investigation updates.
- Supervisors who need to review cases and monitor authorized activity.

## Why is this problem important?

A clear digital workflow can reduce confusion, improve information organization, make case progress easier to follow, and create a more traceable record of important authorized actions.

# 2. Proposed Solution

PoliceProof is a secure, role-based web platform for managing the journey of a police complaint. A citizen submits a complaint and receives a unique Case ID. Authorized police staff can receive, assign, investigate, update, and review the case. Evidence activity is organized into a history, while AI can provide assistive suggestions such as complaint category or priority. Citizens can see permitted status updates without accessing sensitive investigation information.

## How does it solve the problem?

1. Citizen submits the complaint with incident details.
2. PoliceProof creates a unique Case ID.
3. An authorized officer receives and is assigned the case.
4. The officer updates investigation status and records relevant information.
5. Evidence is uploaded and important actions are recorded in an activity/history log.
6. AI suggests category and priority; the authorized officer reviews the suggestion.
7. A supervisor reviews the case before the next authorized step.
8. The citizen views permitted progress updates until the case is closed/resolved.

## What makes the solution useful?

- One connected workflow for complaint, case, and evidence records.
- Unique Case IDs make cases easy to reference.
- Role-based access separates citizen, officer, investigator, supervisor, and administrator views.
- AI is an assistant, not an automatic decision-maker.
- Evidence history records important authorized activity.

# 3. Target Users

| User     | Use of PoliceProof                                        |
|----------|-----------------------------------------------------------|
| Citizens | Submit complaints and view permitted case-status updates. |

| User                  | Use of PoliceProof                                                                  |
|-----------------------|-------------------------------------------------------------------------------------|
| Police Officers       | Receive assigned cases, update investigation progress, and manage case information. |
| Investigators         | Organize investigation information and evidence activity.                           |
| Supervisors           | Review case summaries, evidence history, and investigation progress.                |
| System Administrators | Manage users, roles, permissions, and settings.                                     |

## 4. Key Features

| Feature                 | Purpose                                                      |
|-------------------------|--------------------------------------------------------------|
| User registration/login | Secure access with role-based permissions.                   |
| Complaint submission    | Incident type, location, description, and optional media.    |
| Automatic Case ID       | Unique reference such as KHI-10245.                          |
| Police dashboard        | View cases, status, type, location, and priority.            |
| Case timeline           | Submitted → Assigned → Investigation → Review → Resolved.    |
| AI assistant            | Suggest category and basic priority for officer review.      |
| Evidence management     | Upload permitted evidence and related metadata/history.      |
| Activity / audit log    | Record important authorized actions with user and timestamp. |
| Supervisor review       | Consolidated view for authorized review.                     |
| Citizen tracking        | Show permitted case progress.                                |
| Notifications           | Notify users about permitted status changes.                 |
| Search & filters        | Find cases by ID, type, status, date, or priority.           |

## 5. Business or Wellness Value

PoliceProof is primarily a general wellness/public-service idea. It supports a clearer, more organized, and more accountable public-safety process, while also providing operational value to the organization using it.

### How does it improve people's well-being?

- Citizens can build greater trust in police through a clear and traceable case process.
- Citizens can independently track their case progress through permitted status updates.
- They can know whether their complaint/FIR has been submitted and recorded in the system.
- They can see whether their case is moving through the authorized workflow and being reviewed.
- Clear updates and activity records can improve transparency, accountability, and confidence.

### Who benefits?

- Citizens
- Police officers and investigators
- Police supervisors
- The wider community through a more organized public-service workflow.

## 6. Feasibility

PoliceProof is technically feasible as a university-level MVP using currently available web and AI technologies. The first version should focus on complaint submission, authentication, case assignment, status tracking, evidence metadata, and dashboards. Advanced AI and computer vision can be added after the core workflow works reliably.

### Possible technology stack

| Layer           | Possible Technology             | Use                                    |
|-----------------|---------------------------------|----------------------------------------|
| Frontend        | React / Next.js                 | Citizen portal and dashboards          |
| Backend         | Node.js + Express               | APIs, authentication, business logic   |
| Database        | MongoDB or PostgreSQL           | Users, cases, statuses, metadata, logs |
| Authentication  | JWT + secure cookies/sessions   | Protected role-based access            |
| AI              | Python service / AI API         | Complaint classification assistance    |
| Computer Vision | OpenCV / object-detection model | Optional visual evidence assistance    |
| Storage         | Secure object/file storage      | Images and video evidence              |

### Resources required

- Development laptop and internet.
- Frontend, backend, database, and authentication tools.
- Secure prototype storage for evidence.
- AI/computer-vision service or model if implemented.
- Testing users and fictional/sample cases.

### Challenges

- Privacy and sensitive information.
- Strict role-based access.
- AI/computer-vision errors; outputs must be human-reviewed.
- Secure evidence storage and large video files.
- Official police-system integration would require authorization and technical requirements.

## 7. UI / Design / Diagrams

The assignment specifically asks for visuals where they help explain the idea. The character-based storyboard below demonstrates the complete PoliceProof user journey using fictional characters and sample data.

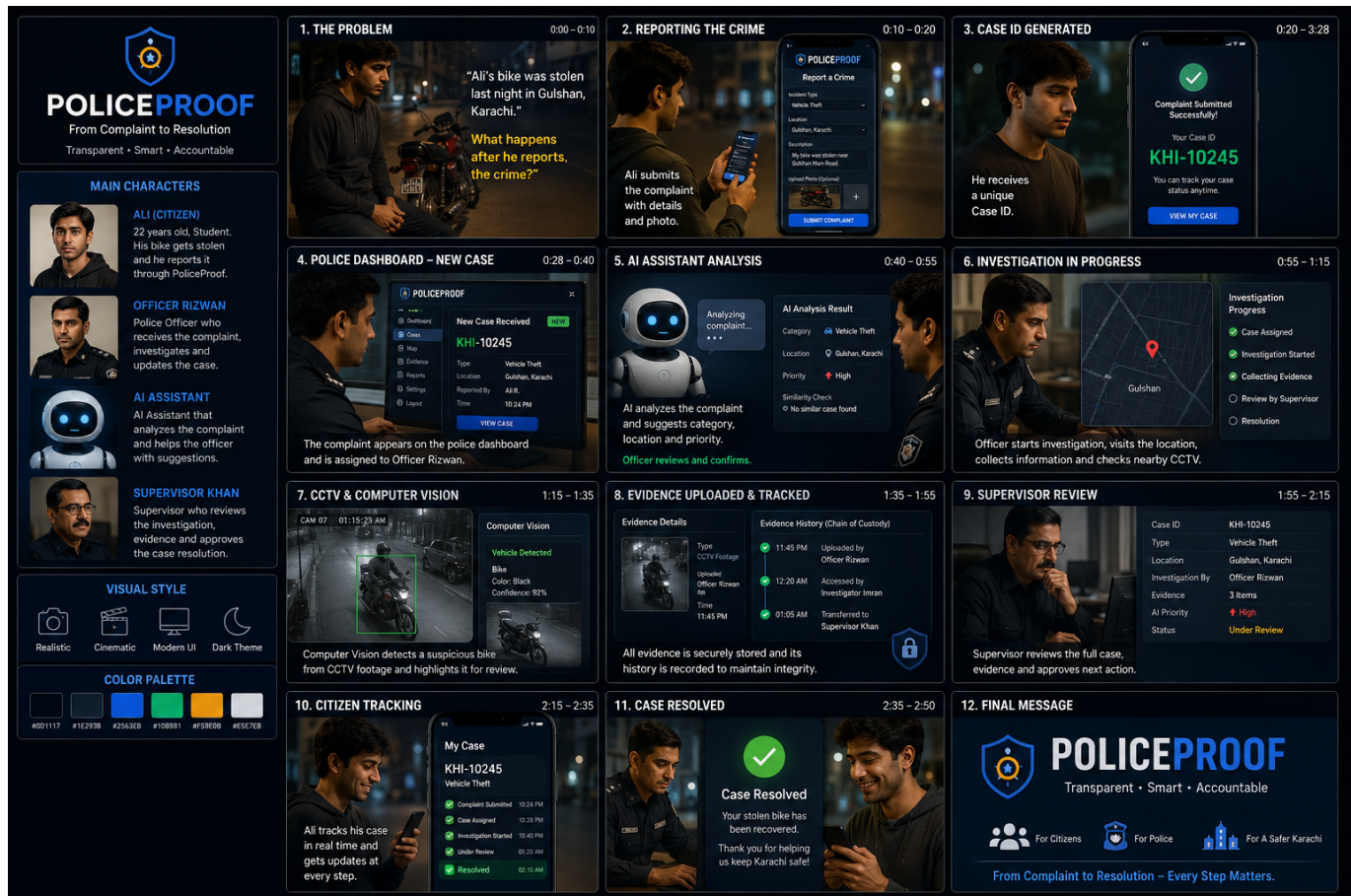


Figure 1 — PoliceProof character storyboard and product journey.

### User Journey

| Citizen                 | PoliceProof        | Police Officer | AI Assistant               | Supervisor                    |
|-------------------------|--------------------|----------------|----------------------------|-------------------------------|
| Reports crime           | Creates Case ID    | Receives case  | Suggests category/priority | Reviews case                  |
| Tracks permitted status | Stores case record | Investigates   | Assists analysis           | Reviews evidence/history      |
| Receives final update   | Updates workflow   | Updates status | Assists only               | Approves next authorized step |

## 8. Final Submission Checklist

| Assignment Requirement                 | Included in PoliceProof                                         |
|----------------------------------------|-----------------------------------------------------------------|
| Name / Roll Number / LinkedIn / GitHub | Placeholders provided on cover page.                            |
| Problem Statement                      | Problem, affected users, and importance explained.              |
| Proposed Solution                      | Product and workflow explained.                                 |
| Target Users                           | Citizens, officers, investigators, supervisors, administrators. |

| Assignment Requirement     | Included in PoliceProof                                                   |
|----------------------------|---------------------------------------------------------------------------|
| Key Features               | Relevant MVP features listed.                                             |
| Business or Wellness Value | Public-service/wellness and operational value explained.                  |
| Feasibility                | Technology, resources, development possibility, and challenges explained. |
| UI / Diagrams              | Character storyboard and user-journey diagram included.                   |
| Professional Formatting    | Consistent headings, tables, colors, and page design.                     |

## 9. Conclusion

PoliceProof is a practical new software-product concept that fits the assignment because it identifies a genuine real-world need and proposes a feasible technology-based MVP. Its main purpose is to create a clear digital journey for a complaint while helping authorized police personnel organize cases, investigation updates, and evidence activity. The concept combines case tracking, role-based access, evidence history, AI assistance, and permitted citizen visibility in one system.

### **POLICEPROOF**

#### **From Complaint to Resolution — Every Step Matters.**

Note: Use fictional/sample cases and data for the university prototype unless official authorization and real requirements/data are provided. AI and computer-vision outputs should remain assistive and human-reviewed.